Prithu Paresh Das

+91 6364811916 | paresh.prithu2205@gmail.com | Prithu Paresh | sawbl4de

OBJECTIVE

To pursue the MS degree in Systems and Control at TU Delft to strengthen my foundations in learning-based control and uncertainty-aware decision-making. I aim to develop autonomous physical systems that integrate real-time perception, model-based control and predictive reasoning to operate safely and reliably under real-world uncertainty.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani — Hyderabad, India
Bachelor of Engineering in Mechanical Engineering & Minor in Data Science (GPA: 8.14/10, MGPA: 8.72/10) — Oct '22 - May '26
- **Relevant Coursework:** Vibrations and Control, Artificial Intelligence, Machine Learning, Advanced Mechanics of Solids

RESEARCH EXPERIENCE

Indian Institute of Science (IISc), Bangalore — Bengaluru, Karnataka
Research Intern - Robotics and Autonomous Systems — June '25 - Present

Optical Ultrafast Spectroscopy and Understanding Material Synthesis (OUSUMS) Lab
- Supervised by Prof. Akshay Singh, Department of Physics
- Developed an **image-based visual servoing** pipeline for a Mitsubishi SCARA to enable **micron-level manipulation** and stacking of 2D materials to build heterostructures
- Replaced open-loop motion with continuous IBVS feedback, eliminating cumulative image-space tracking error and achieving **1–2 micron accuracy**
- Reduced transfer time from **20 minutes to 2 minutes** and executed the first **semi-automated heterostructure assembly**

Data Augmented Control of Autonomous Systems (DACAS) Lab
- Supervised by Prof. Jishnu Keshavan, Department of Mechanical Engineering
- Performed dynamic model identification and implemented an **inverse-dynamics torque controller** on a 6-DOF PiPER manipulator, enabling model-consistent torque commands
- Identified failures due to modeling errors, **reconstructed** and validated the robot's kinematic and dynamic model by correcting joint axes, inertial properties, and frame conventions in the URDF
- Developed safe stiffness-damping profiles for manipulation under dynamic coupling and external disturbances
- Performed joint-level impedance characterization under varying loads, revealing physical interaction limits necessary for safe torque control
- Developing a real-time image-based visual servoing framework for gesture-based control and ball-catching tasks

Birla Institute of Technology and Science (BITS), Pilani — Hyderabad, India
Undergraduate Researcher - Smart Materials and Vibration Attenuation — Jan '24 - Present

NiTi-Based Auxetic Structures for Vibration Mechanics
- Conducting long-term research under Prof. Amit Kumar Gupta on **NiTi-based auxetic structures** for vibration attenuation
- Designed and fabricated re-entrant and bow-tie auxetic geometries using laser machining, EDM, and high-speed machining
- Performed advanced material characterization: microstructure analysis, FE-SEM, XRD, DSC to study phase transformations and mechanical behavior
- Developed an electrodynamic shaker–based setup and demonstrated **40% reduction** in transmitted vibrations in auxetics, relative to solid plates
- Led testing and analysis for two conference papers (**COPEN-13, VETOMAC 2025**) and contributed to a journal manuscript currently being prepared for submission to a Q1 journal
- Contributed to a **Patent** on SMA-based auxetic structures by designing and developing the proof-of-concept prototype

Shape Memory Alloy-Based Actuation Mechanisms
- Developed an automated SMA-driven actuation mechanism for Louvre flaps, for potential use in valve-based flow control, automatic curtain flaps, etc
- Built an spring coiling mechanism to manufacture **custom SMA springs**. Fabricated 0.5mm and 1 mm NiTi springs, characterized thermomechanical response, and integrated into a **working actuator prototype**
- Contributed to patentable concepts and demonstrated application-driven SMA design understanding and capability

Safety, Reliability And Maintainability Centre, CEMILAC — Bengaluru, Karnataka
Research Intern - Reliability, Safety and Airworthiness Analysis — May '24 - July '24
- Undertook an academic research project at the premier national aerospace laboratory responsible for airworthiness certification and safety evaluation
- Developed a novel probabilistic method to estimate the **Reliability** and **Unavailability of One-Shot Applications**
- Validated the developed method for a Payload Delivery System using ISOGRAPH and Monte Carlo simulations
- Executed full-scale **System Safety Analysis**, **FMECA, PHA and Risk Assessment** on the **Landing Gear** unit of an aircraft & other aviation hardware under Directorate General of Civil Aviation standards

PUBLICATIONS AND PATENTS

Conference Papers

- ***Damping Characteristics of Auxetic Structures Made of Shape Memory Alloys*** *- 20th International Conference on Vibration Engineering & Technology Of Machinery, IIT Guwahati Dec 2025* (Accepted)
- ***Estimating the Vibration Attenuation Characteristics of Re-Entrant Bow-Tie Shaped Auxetic Structures through Experimental and Computational Analysis*** *- 13th International Conference on Precision, Meso, Micro and Nano Engineering, NIT Calicut Dec 2024* (Presented)

Journal Paper (In Preparation)

NiTi-Based Auxetic Structures for Enhanced Vibration Damping: Design, Mechanics, and Performance Evaluation
– Targeted for the Journal of Sound and Vibration

Patent

Contributed to Indian Patent Application No. 202311076100: "Shape Memory Alloy Based Auxetic Structure for Damping Applications"

PROJECT HIGHLIGHTS

Decision-Making in Uncertain Environments using LLMs and Bayesian Inference Feb '25 - May '25

- Developed an integrated AI system combining LLM-powered Tic-Tac-Toe (Mistral-7B, Falcon-7B) with a Bayesian Wumpus World agent for **risk-aware decision-making**.
- Engineered an LLM-based move generation pipeline & implemented Bayesian Networks using pgmpy for **reasoning & hazard estimation**. Designed an **adaptive AI strategy** where outcomes dynamically influence the agent's decision-making.

Digital Twin of a Single-Stage Gearbox Jan '25 - May '25

- Fabricated a physical single-stage gearbox using laser cutting, gear cutting, milling, welding, and lathe operations
- Developed a detailed CAD model of the gearbox system in SolidWorks and Fusion 360, ensuring precise component representation and assembly accuracy
- Integrated vibration and acoustic sensors for **real-time monitoring** and data collection
- Developed the complete digital twin by integrating the physical model with a simulation environment using MATLAB and COMSOL to enable real-time performance analysis and predictive maintenance

Condition Monitoring using ML and Vibrational Analysis Jan '25 - March '25

- Developed a **Physics-Informed Neural Network** (PINN) for fault detection and classification in mechanical components based on structural vibration responses
- Lead data procurement, experimental setup design, and model architecture in collaboration with CSIS faculty & students
- Integrated fracture mechanics and vibration physics into the neural network to enhance fault prediction accuracy

Other Projects:
Power Consumption Model using LSTMs | SMA Spring Coiling Mechanism | Development of a SCARA Robot | Website for Grandfather's Poems | Study of Hydrogen Storage Techniques

WORKSHOPS AND CERTIFICATIONS

ADAS/Intelligent Systems Workshop - Dept. of Mechanical Engineering, BITS Pilani; WILP, BITS Pilani Nov '24

- Gained industry insights into ADAS and intelligent control systems, focusing on automation, sensor fusion, and autonomy.

Supervised Machine Learning: Regression and Classification Oct '24

- Stanford Online, DeepLearning.AI - Credential ID D24TG6FFZHTR

Navigating the Pathways of Publishing in High-Quality Journals Aug '24

- Gained insights into research workflow, journal selection, and publishing strategies to improve article acceptance.

TECHNICAL SKILLS

- **Robotics and Simulation:** ROS2, Gazebo, Pinocchio, COMSOL, Ansys. Worked with Mitsubishi SCARA 3CRH and AgileX PiPER 6-DOF platforms
- **Programming and ML:** Python, TensorFlow, Keras, Scikit-learn, LaTex, GIT, SQL
- **Modelling and CAD:** Creo, Fusion, AutoCAD
- **Fabrication & Experimentation:** Laser Machining, 3D Printing, Electrodynamic Shaker, High-Speed Machining

AWARDS AND ACHIEVEMENTS

- Raising A Mathematician Training Program - Selected among **top 100** students of the country to be trained in Advanced Mathematics at a residential camp in Mumbai - May '18
- 1st Dan Black Belt in Okinawa Shorin-Ryu Shorin-Kan Karate, with **12+ years** of practice. Trained and prepared over 50 students for regional and state tournaments